

ASSIGNMENT-2

LINEAR REGRESSION

Linear regression predicts continuous values using straight-line relationship between variables.

Examples of Linear Regression

1. Predicting Crop Yield

Linear regression can estimate crop yield based on factors like rainfall, fertilizer usage, and temperature.

Example: More rainfall and proper fertilizer use can increase crop yield.

2. Estimating Manufacturing Cost

It is used to predict production cost based on number of units produced.

Example: If producing more units increases cost, the model finds that relationship and predicts total cost.

3. Estimating Time

Linear regression can estimate time required to complete a task based on workload.

Example: More tasks or distance leads to more time taken.

4. Estimating Fuel Consumption

It predicts fuel usage based on distance traveled or speed.

Example: As distance increases, fuel consumption also increases in a linear pattern.